

PrimeTrade.ai — Executive Summary

Bitcoin Sentiment × Trader Behaviour Analysis

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1 Objective

The objective of this analysis was to evaluate whether Bitcoin market sentiment—specifically the Fear & Greed Index—meaningfully influences trader behaviour and performance on the Hyperliquid decentralized exchange.

Rather than only describing the data, the analysis aimed to uncover actionable patterns: whether traders perform differently during Fear versus Greed periods, how leverage impacts profitability, and whether meaningful trader archetypes can be identified to inform smarter trading strategies and platform improvements.

2 Dataset Overview

The analysis combines two datasets:

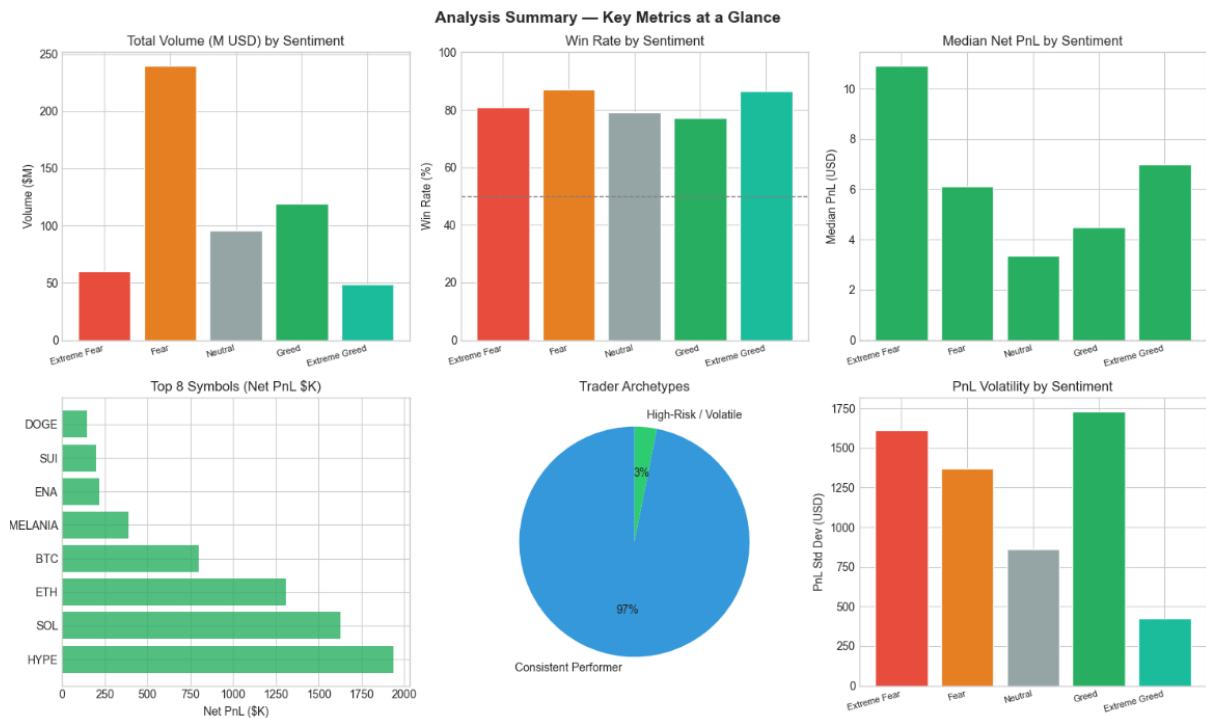
- **Bitcoin Fear & Greed Index:** 2,644 daily sentiment observations from February 2018 to mid-2025, categorised from Extreme Fear to Extreme Greed.
- **Hyperliquid Trade History:** 211,224 individual trades across 32 trader wallets spanning approximately April 2023 to June 2025, including trade side, position size, realised PnL, and fees.

3 Methodology

The analysis followed a structured workflow:

1. **Data Cleaning & Alignment:** Standardised timestamps and aligned trades with daily sentiment data.
2. **Dataset Merging:** Merged trade activity with sentiment regimes.
3. **Exploratory Data Analysis (EDA):** Examined profitability, leverage behaviour, trader performance, and sentiment relationships.
4. **Statistical Testing:** Used Kruskal-Wallis and Spearman correlation tests to validate patterns.
5. **Trader Segmentation:** Applied K-Means clustering to identify trader archetypes.
6. **Business Interpretation:** Converted findings into practical recommendations for PrimeTrade.ai.

4 Executive Snapshot



Key Takeaways:

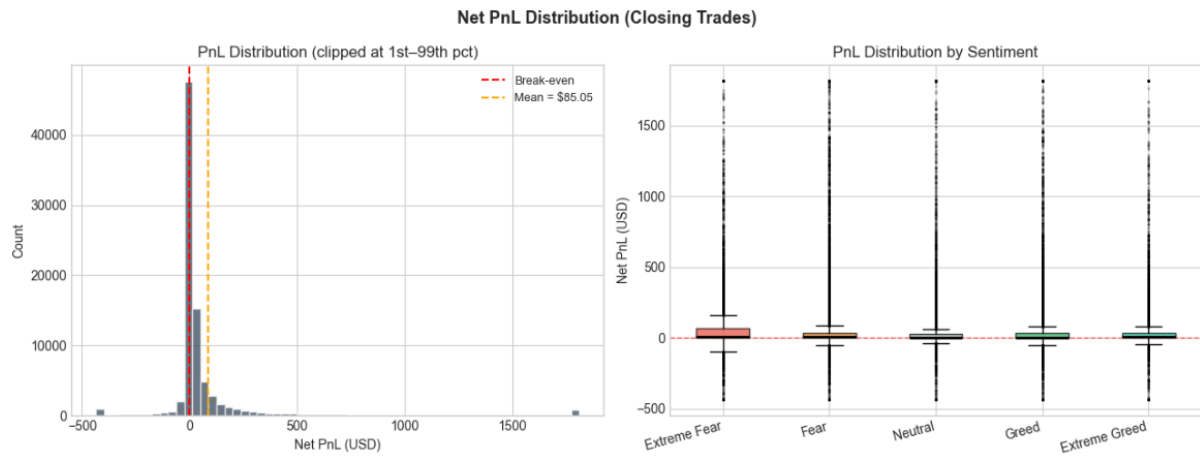
- Fear periods showed wider PnL dispersion, indicating higher uncertainty rather than consistently poor performance.
- Medium leverage ranges appeared to deliver competitive returns with lower volatility compared to extreme leverage.
- Trading activity and profitability were concentrated in a relatively small subset of traders and assets.

5 Top Findings

- **Sentiment Influences Risk More Than Direction:**
Fear periods were associated with wider PnL distributions, suggesting sentiment may act more as a volatility signal than a direct profitability signal.
- **Extreme Leverage Did Not Consistently Improve Outcomes:**
Higher leverage was not strongly associated with stronger returns. Medium leverage categories often produced more stable outcomes.
- **Trader Performance Was Highly Concentrated:**
While many trades were profitable, overall platform performance appeared concentrated among a smaller subset of highly active traders.
- **Asset Concentration Risk Exists:**
HYPE dominated both trading activity and realised PnL, suggesting concentration risk around a small number of symbols.

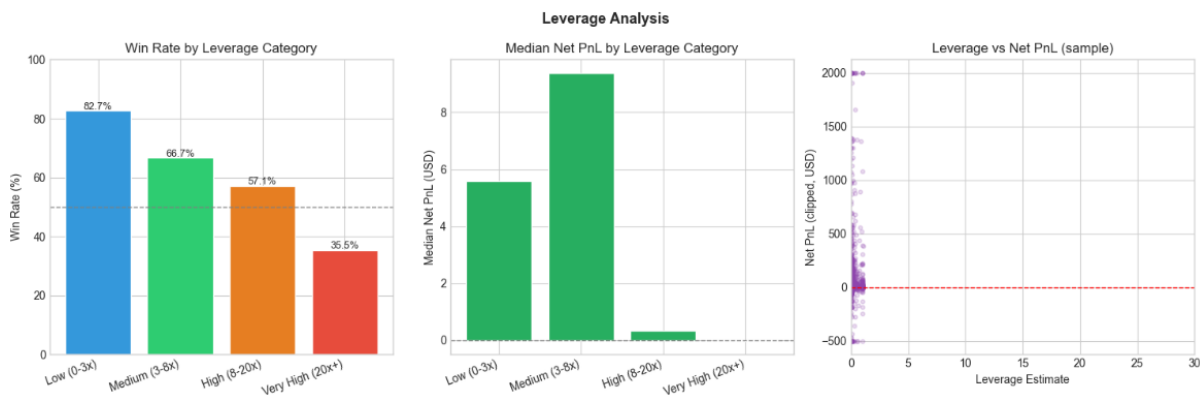
6 Key Visual Insights

Chart 1: Profitability by Sentiment Regime



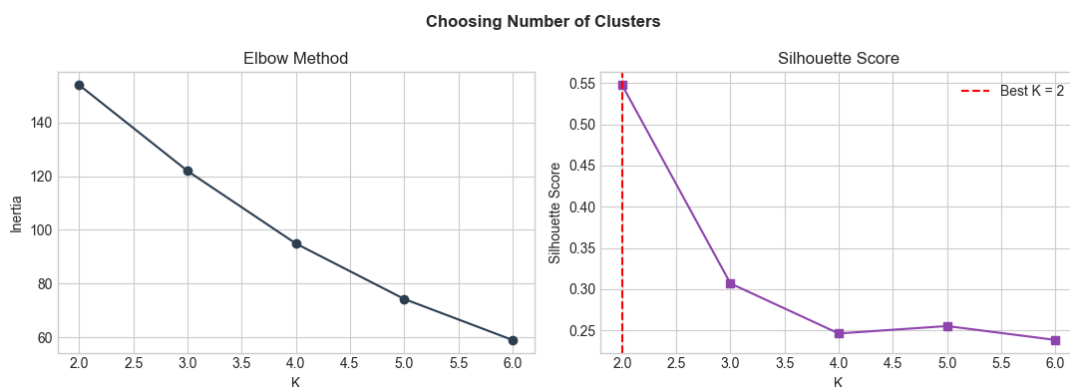
Observation: Fear periods demonstrated wider distributions of profit and loss, whereas Greed periods showed comparatively tighter outcomes.

Chart 2: Leverage Analysis



Observation: Medium leverage ranges showed competitive median PnL while avoiding the higher downside variability observed in extreme leverage groups.

Chart 3: Trader Segmentation



Observation: Trader clustering suggested distinct behavioural profiles, separating more consistent traders from higher-risk profiles.

7 Business Interpretation

The findings suggest that sentiment appears to be a stronger signal for *market uncertainty and volatility* than for direct profitability.

In addition, trader behaviour was highly heterogeneous, meaning aggregate platform metrics may mask meaningful differences in trading style, leverage preference, and consistency.

This creates opportunities for more personalised risk management and trading experiences.

8 Recommendations for PrimeTrade.ai

1. Sentiment-Aware Risk Controls:

Display the Fear & Greed Index directly within the trading interface to provide additional market context.

2. Leverage Defaults Tied to Sentiment:

Consider lowering pre-filled default leverage during Extreme Fear and Extreme Greed periods while preserving trader choice.

3. Behavioural Alerts for At-Risk Traders:

Repeated losses combined with increasing position size may indicate loss-chasing behaviour. Lightweight prompts may encourage healthier risk decisions.

4. Trader Segmentation for Personalisation:

Trader clustering may support personalised leverage defaults, educational content, and trader-specific guidance.

5. Symbol Concentration Monitoring:

Monitor dominant asset concentration to identify potential platform-wide risk exposure.

6. Risk-Adjusted Performance Metrics:

Enhance trader leaderboards using consistency-based metrics alongside total PnL.

9 Limitations

- **Sample Size:** Only 32 trader accounts were available, limiting broad statistical generalisation.
- **BTC-Specific Sentiment Proxy:** The Fear & Greed Index is Bitcoin-focused, while many trades involved altcoins.
- **Leverage Approximation:** Leverage was estimated rather than directly recorded.
- **Market Regime Bias:** The analysed period was predominantly bullish and may not generalise to prolonged bear markets.

10 Final Conclusion

The Fear & Greed Index appears to be more useful as a **contextual risk signal** than a direct trading signal.

While sentiment alone does not determine profitability, it appears to influence trader behaviour through changes in volatility and risk appetite.

Combining sentiment context with trader segmentation may provide PrimeTrade.ai with opportunities to improve risk awareness, personalise trader experiences, and support more sustainable trading behaviour.